- Step 2. Simplify the expressions on both sides of the equation.
- Step 3. Determine whether the resulting equation is true.
  If it is true, the number is a solution.
  If it is not true, the number is not a solution.

- **Properties of Equalities**

Subtraction Property of Equality	Addition Property of Equality
For any numbers a, b, c, if $a = b$ then $a - c = b - c$.	For any numbers a, b, c, if $a = b$ then $a + c = b + c$.

- **Division Property of Equality**
  - For any numbers a, b, c, and $c \neq 0$
    If $a = b$, then $\frac{a}{c} = \frac{b}{c}$.

Exercises

Review Exercises

Introduction to Integers

Locate Positive and Negative Numbers on the Number Line

In the following exercises, locate and label the integer on the number line.

353. 5 **354.** -5 **355.** -3

356. 3 **357.** -8 **358.** -7

Order Positive and Negative Numbers

In the following exercises, order each of the following pairs of numbers, using $<$ or $>$.

359. 4__8 **360.** -6__3 **361.** -5__-10

362. -9__-4 **363.** 2__-7 **364.** -3__1

Find Opposites

In the following exercises, find the opposite of each number.

365. 6 **366.** -2 **367.** -4

368. 3

In the following exercises, simplify.

369. (a) $-(8)$ (b) $-(-8)$ **370.** (a) $-(9)$ (b) $-(-9)$

In the following exercises, evaluate.

371. $-x$, when
(a) $x = 32$
(b) $x = -32$

372. $-n$, when
(a) $n = 20$
(a) $n = -20$

Simplify Absolute Values

In the following exercises, simplify.

373. $|-21|$ **374.** $|-42|$ **375.** $|36|$

376. $-|15|$ **377.** $|0|$ **378.** $-|-75|$

In the following exercises, evaluate.

379. $|x|$ when $x = -14$

380. $-|r|$ when $r = 27$

381. $-|-y|$ when $y = 33$

382. $|-n|$ when $n = -4$

In the following exercises, fill in $<$, $>$, or $=$ for each of the following pairs of numbers.

383. $-|-4|$ __ 4

384. -2 __ $|-2|$

385. $-|-6|$ __ -6

386. $-|-9|$ __ $|-9|$

In the following exercises, simplify.

387. $-(-55)$ and $-|-55|$

388. $-(-48)$ and $-|-48|$

389. $|12 - 5|$

390. $|9 + 7|$

391. $6\,|-9|$

392. $|14-8| - |-2|$

393. $|9 - 3| - |5 - 12|$

394. $5 + 4\,|15 - 3|$

Translate Phrases to Expressions with Integers

In the following exercises, translate each of the following phrases into expressions with positive or negative numbers.

395. the opposite of 16

396. the opposite of -8

397. negative 3

398. 19 minus negative 12

399. a temperature of 10 below zero

400. an elevation of 85 feet below sea level

Add Integers

Model Addition of Integers

In the following exercises, model the following to find the sum.

401. $3 + 7$

402. $-2 + 6$

403. $5 + (-4)$

404. $-3 + (-6)$

Simplify Expressions with Integers

In the following exercises, simplify each expression.

405. $14 + 82$

406. $-33 + (-67)$

407. $-75 + 25$

408. $54 + (-28)$

409. $11 + (-15) + 3$

410. $-19 + (-42) + 12$

411. $-3 + 6(-1 + 5)$

412. $10 + 4(-3 + 7)$

Evaluate Variable Expressions with Integers

In the following exercises, evaluate each expression.

413. $n + 4$ when
ⓐ $n = -1$
ⓑ $n = -20$

414. $x + (-9)$ when
ⓐ $x = 3$ ⓑ $x = -3$

415. $(x + y)^3$ when $x = -4, y = 1$

416. $(u + v)^2$ when $u = -4, v = 11$

Translate Word Phrases to Algebraic Expressions

In the following exercises, translate each phrase into an algebraic expression and then simplify.

417. the sum of -8 and 2

418. 4 more than -12

419. 10 more than the sum of -5 and -6

420. the sum of 3 and -5, increased by 18

Add Integers in Applications

In the following exercises, solve.

421. **Temperature** On Monday, the high temperature in Denver was -4 degrees. Tuesday's high temperature was 20 degrees more. What was the high temperature on Tuesday?

422. **Credit** Frida owed \$75 on her credit card. Then she charged \$21 more. What was her new balance?

Subtract Integers

Model Subtraction of Integers

In the following exercises, model the following.

423. $6 - 1$

424. $-4 - (-3)$

425. $2 - (-5)$

426. $-1 - 4$

Simplify Expressions with Integers

In the following exercises, simplify each expression.

427. $24 - 16$

428. $19 - (-9)$

429. $-31 - 7$

430. $-40 - (-11)$

431. $-52 - (-17) - 23$

432. $25 - (-3 - 9)$

433. $(1 - 7) - (3 - 8)$

434. $3^2 - 7^2$

Evaluate Variable Expressions with Integers

In the following exercises, evaluate each expression.

435. $x - 7$ when
ⓐ $x = 5$ ⓑ $x = -4$

436. $10 - y$ when
ⓐ $y = 15$
ⓑ $y = -16$

437. $2n^2 - n + 5$ when $n = -4$

438. $-15 - 3u^2$ when $u = -5$

Translate Phrases to Algebraic Expressions

In the following exercises, translate each phrase into an algebraic expression and then simplify.

439. the difference of -12 and 5

440. subtract 23 from -50

Subtract Integers in Applications

In the following exercises, solve the given applications.

441. **Temperature** One morning the temperature in Bangor, Maine was 18 degrees. By afternoon, it had dropped 20 degrees. What was the afternoon temperature?

442. **Temperature** On January 4, the high temperature in Laredo, Texas was 78 degrees, and the high in Houlton, Maine was -28 degrees. What was the difference in temperature of Laredo and Houlton?

Multiply and Divide Integers

Multiply Integers

In the following exercises, multiply.

443. $-9 \cdot 4$

444. $5(-7)$

445. $(-11)(-11)$

446. $-1 \cdot 6$

Divide Integers

In the following exercises, divide.

447. $56 \div (-8)$

448. $-120 \div (-6)$

449. $-96 \div 12$

450. $96 \div (-16)$

451. $45 \div (-1)$

452. $-162 \div (-1)$

Simplify Expressions with Integers

In the following exercises, simplify each expression.

453. $5(-9) - 3(-12)$

454. $(-2)^5$

455. -3^4

456. $(-3)(4)(-5)(-6)$

457. $42 - 4(6 - 9)$

458. $(8 - 15)(9 - 3)$

459. $-2(-18) \div 9$

460. $45 \div (-3) - 12$

Evaluate Variable Expressions with Integers

In the following exercises, evaluate each expression.

461. $7x - 3$ when $x = -9$

462. $16 - 2n$ when $n = -8$

463. $5a + 8b$ when $a = -2, b = -6$

464. $x^2 + 5x + 4$ when $x = -3$

Translate Word Phrases to Algebraic Expressions

In the following exercises, translate to an algebraic expression and simplify if possible.

465. the product of -12 and 6

466. the quotient of 3 and the sum of -7 and s

Solve Equations using Integers; The Division Property of Equality

Determine Whether a Number is a Solution of an Equation

In the following exercises, determine whether each number is a solution of the given equation.

467. $5x - 10 = -35$

ⓐ $x = -9$ ⓑ $x = -5$

ⓒ $x = 5$

468. $8u + 24 = -32$

ⓐ $u = -7$ ⓑ $u = -1$

ⓒ $u = 7$

Using the Addition and Subtraction Properties of Equality

In the following exercises, solve.

469. $a + 14 = 2$

470. $b - 9 = -15$

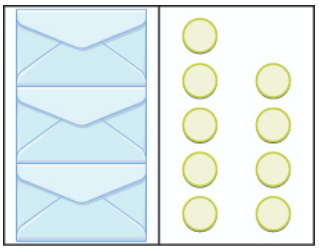
471. $c + (-10) = -17$

472. $d - (-6) = -26$

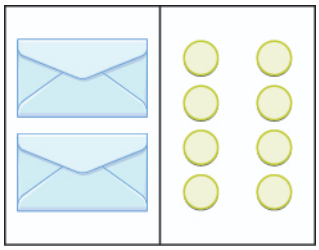
Model the Division Property of Equality

In the following exercises, write the equation modeled by the envelopes and counters. Then solve it.

473.



474.



Solve Equations Using the Division Property of Equality

In the following exercises, solve each equation using the division property of equality and check the solution.

475. $8p = 72$

476. $-12q = 48$

477. $-16r = -64$

478. $-5s = -100$

Translate to an Equation and Solve.

In the following exercises, translate and solve.

479. The product of -6 and y is -42

480. The difference of z and -13 is -18.

481. Four more than m is -48.

482. The product of -21 and n is 63.

Everyday Math

483. Describe how you have used two topics from this chapter in your life outside of your math class during the past month.

Practice Test

484. Locate and label $0, 2, -4,$ and -1 on a number line.

In the following exercises, compare the numbers, using $<$ or $>$ or $=$.

485. ⓐ $-6 __ 3$
ⓑ $-1 __ -4$

486. ⓐ $-5 __ |-5|$
ⓑ $-|-2| __ -2$

In the following exercises, find the opposite of each number.

487. ⓐ -7 ⓑ 8

In the following exercises, simplify.

488. $-(-22)$

489. $|4 - 9|$

490. $-8 + 6$

491. $-15 + (-12)$

492. $-7 - (-3)$

493. $10 - (5 - 6)$

494. $-3 \cdot 8$

495. $-6(-9)$

496. $70 \div (-7)$

497. $(-2)^3$

498. -4^2

499. $16 - 3(5 - 7)$

500. $|21 - 6| - |-8|$

In the following exercises, evaluate.

501. $35 - a$ when $a = -4$

502. $(-2r)^2$ when $r = 3$

503. $3m - 2n$ when $m = 6,\ n = -8$

504. $-|-y|$ when $y = 17$

In the following exercises, translate each phrase into an algebraic expression and then simplify, if possible.

505. the difference of −7 and −4

506. the quotient of 25 and the sum of m and n.

In the following exercises, solve.

507. Early one morning, the temperature in Syracuse was $-8°$F. By noon, it had risen $12°$. What was the temperature at noon?

508. Collette owed $\$128$ on her credit card. Then she charged $\$65$. What was her new balance?

In the following exercises, solve.

509. $n + 6 = 5$

510. $p - 11 = -4$

511. $-9r = -54$

In the following exercises, translate and solve.

512. The product of 15 and x is 75.

513. Eight less than y is -32.